

Practical Workbook
CT-361
Artificial Intelligence & Expert Systems



Name: Moazzam Farooqui

Roll No: CT-24068

Batch: 2024

Department: CSIT

Department of Computer Science & Information Technology
NED University of Engineering & Technology

LAB # 02

DATA OBJECTS IN PROLOG

Q1. You are required to build a University Knowledge Base in Prolog. Your program must include all four Prolog data objects:

- 1. Atoms → for names, cities, and course codes.**
- 2. Numbers → for ages, grades, and course credits.**
- 3. Variables → to allow queries with unknown values.**
- 4. Structures → to represent complete information about students, courses, teachers, and grades.**

1. Define the Knowledge Base (Facts)

•Create facts for **students** in the form:
student(Name, Age, Department).

•Create facts for **courses** in the form:
course(Code, Title, Credits).

•Create facts for **teachers** in the form:
teacher(Name, Department).

•Create facts for **grades** in the form:
grade(StudentName, CourseCode, Marks).

•Add facts that show which **teacher teaches which course**:
teaches(TeacherName, CourseCode).

2. Queries

1.The department and age of a student.

2.The marks of a student in a particular course.

3.The teacher who teaches a given course.

4.The list of all students in a given department.

5.The list of all courses with their credits.

6.Which students scored above 80 in any course.

7.Which teacher teaches Ali's course.

8.Who are Ali's classmates (students enrolled in the same course).

SOLUTION:

% Q1

student(mudassir,20,cs).

student(mazz,21,cs).

student(saud,22,ee).

student(ali,20,math).

course(english,functionalenglish,3).

course(maths,linear_algebra,4).

course(science,applied_physics,3).

course(islamiat,islamic_studies,2).

teacher(ms_ayesha,cs).

teacher(ms_rahela,ee).

teacher(ms_samia,math).

grade(mudassir,english,85).

grade(mazz,english,92).

grade(saud,science,78).

grade(mudassir,maths,75).

grade(ali,english,80).

teaches(ms_ayesha,english).

teaches(ms_ayesha,maths).

teaches(ms_rahela,science).

OUTPUT:

```
105 ?- student(ali, Age, Department).
```

```
Age = 20,
```

```
Department = math.
```

```
106 ?- grade(ali, english, Marks).
```

```
Marks = 80.
```

```
107 ?- teaches(TeacherName, english).
```

```
TeacherName = ms_ayesha.
```

```
108 ?- student(StudentName, _, cs).
```

```
StudentName = mudassir ;
```

```
StudentName = mazz.
```

```
109 ?- course(CourseCode, _, Credits).
```

```
CourseCode = english,
```

```
Credits = 3 ;
```

```
CourseCode = maths,
```

```
Credits = 4 ;
```

```
CourseCode = science,
```

```
Credits = 3 ;
```

```
CourseCode = islamiat,
```

```
Credits = 2.
```

```
110 ?- grade(StudentName, _, Marks), Marks > 80.
```

```
StudentName = mudassir,
```

```
Marks = 85 ;
```

```
StudentName = mazz,
```

```
Marks = 92 ;
```

```
false.
```

```
111 ?- grade(ali, CourseCode, _), teaches(TeacherName, CourseCode).
```

```
CourseCode = english,
```

```
TeacherName = ms_ayesha.
```

```
112 ?- grade(ali, CourseCode, _), grade(ClassmateName, CourseCode, _), ClassmateName \= ali.
```

```
CourseCode = english,
```

```
ClassmateName = mudassir ;
```

```
CourseCode = english,
```

```
ClassmateName = mazz ;
```

```
false.
```

**Q2. You are required to build a Hospital Knowledge Base in Prolog.
Your program must include all four Prolog data objects:**

- 1. Atoms → for names, diseases, and departments.**
- 2. Numbers → for ages, room numbers, and treatment costs.**
- 3. Variables → to allow queries with unknown values.**
- 4. Structures → to represent complete information about patients, doctors, diseases, and treatments.**

1. Define the Knowledge Base (Facts)

•Create facts for patients in the form:
patient(Name, Age, Disease).

•Create facts for doctors in the form:
doctor(Name, Department).

•Create facts for rooms in the form:
room(Number, Type).

•Create facts for treatments in the form:
treatment(Disease, Medicine, Cost).

•Add facts that show which doctor treats
which disease:
treats(DoctorName, Disease).

2. Queries

1.The age and disease of a particular patient.

1.The department of a doctor.

2.The doctor who treats a given disease.

3.The list of all patients suffering from a given disease.

4.The room number and type assigned to a patient.

5.Which treatments cost more than 5000.

6.Which doctor treats Ahmed's disease.

7.Who are Fatima's fellow patients (patients suffering from the same disease).

SOLUTION:

% Q2

```
patient(ahmed,45,fever).  
patient(fatima,30,cough).  
patient(ali,25,fever).  
patient(sara,50,asthma).
```

```
doctor(dr_sadia,cardiology).  
doctor(dr_saim,general).  
doctor(dr_shahab,skin).
```

```
room(101,icu).  
room(102,general).  
room(103,private).
```

```
treatment(fever,panadol,1500).  
treatment(cough,syrup,6000).  
treatment(asthma,inhaler,8000).
```

```
treats(dr_sadia,fever).  
treats(dr_saim,cough).  
treats(dr_shahab,asthma).
```

```
assigned_room(ahmed,101).  
assigned_room(fatima,102).  
assigned_room(ali,103).
```

OUTPUT:

```
115 ?- patient(ahmed, Age, Disease).  
Age = 45,  
Disease = fever.
```

```
116 ?- doctor(dr_sadia, Department).  
Department = cardiology.
```

```
117 ?- treats(DoctorName, fever).  
DoctorName = dr_sadia.
```

```
118 ?- patient(PatientName, _, fever).  
PatientName = ahmed ;  
PatientName = ali ;  
false.
```

```
119 ?- assigned_room(ahmed, RoomNumber), room(RoomNumber, Type).  
RoomNumber = 101,  
Type = icu.
```

```
120 ?- treatment(Disease, Medicine, Cost), Cost > 5000.  
Disease = cough,  
Medicine = syrup,  
Cost = 6000 ;  
Disease = asthma,  
Medicine = inhaler,  
Cost = 8000.
```

```
121 ?- patient(ahmed, _, Disease), treats(DoctorName, Disease).  
Disease = fever,  
DoctorName = dr_sadia.
```

```
122 ?- patient(fatima, _, Disease), patient(FellowPatient, _, Disease), FellowPatient \= fatima.  
false.
```

Q3. Queries to Ask:

1. Does Ahmed suffer from flu?
2. What disease does Sara have?
3. Which patients are in ward2?
4. Who treats diabetes?
5. Which doctor treats Ahmed's disease?
6. List all patients who have the same disease as Fatima.
7. Which doctors treat patients in ward1?

```
patient(ahmed, flu, ward1).  
patient(sara, diabetes, ward2).  
patient(fatima, flu, ward1).  
patient(ali, heart_disease, icu).  
patient(hassan, diabetes, ward2).
```

```
doctor(khalid, cardiology).  
doctor(samina, general).  
doctor(asad, endocrinology).
```

```
treats(samina, flu).  
treats(asad, diabetes).  
treats(khalid, heart_disease).
```

SOLUTION:

```
% Q3
patient(ahmed,flu,ward1).
patient(sara,diabetes,ward2).
patient(fatima,flu,ward1).
patient(ali,heart_disease,icu).
patient(hassan,diabetes,ward2).

doctor(khalid,cardiology).
doctor(samina,general).
doctor(asad,endocrinology).

treats(samina,flu).
treats(asad,diabetes).
treats(khalid,heart_disease).
```

OUTPUT:

```
133 ?- patient(ahmed,flu,_).
true .

134 ?- patient(sara,Disease,_).
Disease = diabetes.

135 ?- patient(PatientName,_,ward2).
PatientName = sara ;
PatientName = hassan.

136 ?- treats(DoctorName,diabetes).
DoctorName = asad.

137 ?- patient(ahmed,Disease,_),treats(DoctorName,Disease).
Disease = flu,
DoctorName = samina.

138 ?- patient(fatima,Disease,_),patient(PatientName,Disease,_),PatientName \= fatima.
Disease = flu,
PatientName = ahmed ;
false.

139 ?- patient(_,Disease,ward1),treats(DoctorName,Disease).
Disease = flu,
DoctorName = samina ;
Disease = flu,
DoctorName = samina.
```

Q4. Queries to Ask:

- 1. What is Ahmed's salary?**
- 2. Which department does Sara work in?**
- 3. List all employees in the IT department.**
- 4. What projects are handled by the accounts department?**
- 5. Who works on project alpha?**
- 6. Which employees work on the same project as Fatima?**
- 7. Which department is responsible for project gamma?**
- 8. What is the salary of employees working on project alpha?**

```
employee(ali, accounts, 35000).  
employee(sara, hr, 40000).  
employee(ahmed, it, 55000).  
employee(fatima, it, 60000).  
employee(hassan, marketing, 45000).
```

```
project(alpha, it).  
project(beta, accounts).  
project(gamma, marketing).
```

```
works_on(ali, beta).  
works_on(ahmed, alpha).  
works_on(fatima, alpha).  
works_on(hassan, gamma).
```

SOLUTION:

```
% Q4
employee(ali,accounts,35000).
employee(sara,hr,40000).
employee(ahmed,it,55000).
employee(fatima,it,60000).
employee(hassan,marketing,45000).

project(alpha,it).
project(beta,accounts).
project(gamma,marketing).

works_on(ali,beta).
works_on(ahmed,alpha).
works_on(fatima,alpha).
works_on(hassan,gamma).
```

OUTPUT:

```
140 ?- consult(['c:/Users/USER/Desktop/AIES Labs/q4 2.pl']).
true.

141 ?- employee(ahmed,_,Salary).
Salary = 55000.

142 ?- employee(sara,Department,_).
Department = hr.

143 ?- employee(EmployeeName,it,_).
EmployeeName = ahmed ;
EmployeeName = fatima ;
false.

144 ?- project(ProjectName,accounts).
ProjectName = beta.

145 ?- works_on(EmployeeName,alpha).
EmployeeName = ahmed ;
EmployeeName = fatima ;
false.

146 ?- works_on(fatima,ProjectName),works_on(EmployeeName,ProjectName),EmployeeName \= fatima.
ProjectName = alpha,
EmployeeName = ahmed ;
false.

147 ?- project(gamma,Department).
Department = marketing.

148 ?- works_on(EmployeeName,alpha),employee(EmployeeName,_,Salary).
EmployeeName = ahmed,
Salary = 55000 ;
EmployeeName = fatima,
Salary = 60000 ;
false.
```